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The Evolution of Portfolio Theory Under Risk & Uncertainty: From Mean–Variance to AI-Augmented Investing

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ABSTRACT

This article summarizes the evolution of portfolio theory from mean–variance optimization to AI-augmented investment systems. Rather than treating portfolio models as isolated techniques, it organizes the literature as a sequence of responses to different forms of uncertainty: variance, systematic risk, expected-return estimation error, downside risk, tail loss, parameter ambiguity, risk-budgeting instability, conditional volatility, regime dependence, high-dimensional prediction, and AI governance. Combining a systematic literature review, public-metadata bibliometric analysis, and thematic synthesis, the article covers leading academic journals and practitioner-oriented outlets, including JF, JFE, RFS, RF, JFQA, Management Science, FAJ, JPM, FMPM, and JAM. The survey highlights how classical portfolio theory, CAPM and multifactor allocation, Bayesian/Black–Litterman models, CVaR, robust optimization, risk parity, volatility management, regime-switching allocation, parametric portfolio policies, machine learning, and LLM-based agents form a cumulative intellectual architecture.

The article is designed as a compact roadmap for doctoral students, academic researchers, and quantitative researchers in hedge funds and asset management who seek a rapid but rigorous understanding of how portfolio theory has evolved and how its core problems remain relevant in the AI era.

“The evolution of portfolio theory is not a succession of disconnected models, but a cumulative architecture for transforming uncertainty into disciplined investment decisions.”

1 | Introduction

Portfolio theory is one of the central intellectual achievements of modern finance. Its development begins with a simple but profound insight: investment risk should not be evaluated asset by asset, but through the joint behavior of assets inside a

portfolio. Markowitz’s “Portfolio Selection” transformed diversification from an intuitive investment principle into a formal optimization problem in which expected returns, variances, and covariances jointly determine efficient capital allocation (Markowitz 1952). This contribution remains the foundation of modern portfolio theory, but the field has never stood still. Over more than seven decades, portfolio theory has evolved through repeated attempts to address the limitations of earlier models. The survey organizes this evolution into nine generations: mean–variance portfolio theory; CAPM and multifactor allocation; Bayesian portfolio theory and Black–Litterman; downside risk, VaR, and CVaR; robust optimization and the 1/N challenge; risk parity and volatility-managed port-

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folios; regime-switching allocation; parametric portfolio policies and machine-learning allocation; and LLM/AI-augmented investing.

The classical Markowitz framework defines the investor as choosing portfolio weights w to trade off expected return against risk:

$$\max_w w' \mu - \frac{\gamma}{2} w' \Sigma w$$

subject to:

$$1'w = 1.$$

In this formulation, μ denotes the vector of expected returns, Σ denotes the covariance matrix, and γ represents risk aversion. The elegance of this framework lies in its generality. It shows that a security should be held not only because of its own expected return or volatility, but because of how it changes the return-risk profile of the whole portfolio. Yet this same framework also creates the core tensions that later portfolio theory had to confront. Variance treats upside and downside deviations symmetrically; expected returns are difficult to estimate; covariance matrices are unstable in large dimensions; and a static single-period efficient frontier cannot fully capture changing market states, institutional constraints, transaction costs, or the high-dimensional information environment faced by modern investors.

The second generation of portfolio theory emerged from equilibrium asset pricing and multifactor allocation. Sharpe's capital asset pricing model shifted attention from total portfolio variance to systematic market risk and made beta the central measure of priced risk (Sharpe 1964). Together with Lintner (1965), Mossin (1966), Black (1972), and Ross (1976), this literature embedded portfolio choice within an equilibrium theory of expected returns. In its familiar form, the CAPM implies:

$$E(R_i - R_f) = \beta_i E(R_M - R_f)$$

This equation links portfolio theory to asset pricing by showing that only non-diversifiable covariance with the market should command a risk premium. The CAPM therefore recast portfolio theory as a theory of systematic exposure. However, the empirical asset-pricing literature soon showed that market beta alone was insufficient to explain the cross-section of returns. Fama and French's three-factor model introduced size and value as additional common return factors (Fama and French 1993), while Carhart (1997) incorporated momentum into performance evaluation. The evidence on momentum and value strategies further expanded the factor-allocation view of portfolio construction (Jegadeesh and Titman 1993; Asness et al. 2013). As a result, portfolio theory moved from choosing individual securities to choosing exposures to systematic sources of return.

A third generation emerged from the estimation-error problem in expected returns. The Markowitz framework requires investors to estimate μ , but expected returns are much harder to estimate reliably than covariances. This problem motivated Bayesian portfolio theory, which treats expected returns and, in

some cases, covariances as uncertain objects rather than fixed inputs. Klein and Bawa (1976) show that estimation risk affects optimal portfolio choice, while Jorion's Bayes-Stein approach stabilizes expected-return estimates through shrinkage toward a prior or grand mean (Jorion 1986). Pástor (2000), Pástor and Stambaugh (2000), and Avramov and Zhou (2010) further develop Bayesian perspectives on portfolio choice, model uncertainty, and asset-pricing uncertainty.

The Black-Litterman model represents the most influential practitioner-facing version of this Bayesian logic. Instead of asking investors to provide a full vector of expected returns directly, the model begins with equilibrium-implied returns and then incorporates investor views with confidence levels (Black and Litterman 1992). In simplified form, equilibrium-implied excess returns can be written as:

$$\pi = \delta \Sigma w_m,$$

where w_m is the market-capitalization portfolio and δ is a risk-aversion coefficient. Investor views can be expressed as:

$$P\mu = Q + \varepsilon, \varepsilon \sim N(0, \Omega),$$

where P maps assets into views, Q gives the view vector, and Ω captures uncertainty about those views. The posterior expected return vector is then:

$$\mu_{BL} = \left[(\tau \Sigma)^{-1} + P' \Omega^{-1} P \right]^{-1} \left[(\tau \Sigma)^{-1} \pi + P' \Omega^{-1} Q \right].$$

Black-Litterman therefore connects Markowitz optimization, CAPM-style equilibrium, Bayesian updating, and practitioner judgment. Satchell and Scowcroft (2000), He and Litterman (2002), Idzorek (2004), and Meucci (2010) further explain and extend the model for implementation. This makes Bayesian portfolio theory and Black-Litterman a natural bridge between classical mean-variance optimization and later robust optimization. In this generation, uncertainty is no longer only uncertainty about future returns; it is also uncertainty about the expected-return inputs used to construct portfolios.

The fourth generation challenged the use of variance as the dominant risk measure and developed downside-risk and tail-risk approaches. Investors usually do not dislike all volatility equally. They welcome upside variation but dislike losses, drawdowns, funding shortfalls, and returns below a required threshold. This intuition underlies the safety-first and downside-risk traditions. The lower-partial-moment literature reframed risk as shortfall below a target rather than dispersion around a mean (Bawa 1975; Fishburn 1977; Harlow and Rao 1989; Harlow 1991; Sortino and van der Meer 1991). For a portfolio return R_p and target return τ , the lower partial moment of order n is:

$$LPM_n(\tau) = E[\max(\tau - R_p, 0)^n]$$

This formulation is more closely aligned with the objectives of pension funds, insurance companies, endowments, hedge funds, and private wealth managers, whose investment problems are often defined by minimum acceptable returns, liabilities, drawdowns, and benchmark-relative losses. Downside-risk the-

ory therefore expanded portfolio theory by making investor objectives more realistic and asymmetric. It also created a direct bridge between academic finance and practitioner-oriented performance evaluation, where downside deviation and target-based risk are often more intuitive than total volatility.

The institutionalization of financial risk management generated a further shift from downside risk to tail-risk measurement. Value-at-Risk became popular because it offered a simple quantile-based summary of potential loss. However, VaR is theoretically limited because it does not generally satisfy the coherence properties required of a well-behaved risk measure. Artzner et al. (1999) formalized this critique by introducing coherent risk measures. Rockafellar and Uryasev (2000, 2002) then made Conditional Value-at-Risk, or expected shortfall, central to portfolio optimization by showing that it can be minimized through a tractable convex formulation. For a portfolio loss L , CVaR can be written as:

$$CVaR_{\alpha}(L) = \min_{\eta} \left[\eta + \frac{1}{1-\alpha} E(L - \eta)^+ \right].$$

This was a decisive step in the evolution of portfolio theory. Risk was no longer treated as average dispersion or even below-target shortfall, but as the expected severity of losses in the worst states. This made CVaR especially relevant for leveraged strategies, derivatives, credit portfolios, hedge funds, and institutional portfolios exposed to crash risk. It also moved portfolio theory closer to operations research because tail-risk control could now be expressed as an optimization problem rather than only a reporting metric.

A fifth generation addressed a different form of uncertainty: not uncertainty about future returns conditional on known parameters, but uncertainty about the parameters themselves. This shift is central to the practical history of portfolio theory because the Markowitz framework is highly sensitive to estimation errors in expected returns and covariances. Michaud (1989) famously characterized mean-variance optimization as an “error maximizer,” capturing the concern that mathematically optimized portfolios may amplify input noise rather than improve real investment performance. Best and Grauer (1991) and Chopra and Ziemba (1993) show that small errors in means, variances, and covariances can generate large changes in optimal weights. Jagannathan and Ma (2003) further show that even “wrong” portfolio constraints may improve out-of-sample performance because they act as implicit shrinkage devices, while Ledoit and Wolf (2004) address covariance-matrix instability through shrinkage estimation. Kan and Zhou (2007) formalize optimal portfolio choice under parameter uncertainty.

The same concern also explains the enduring importance of the naïve $1/N$ portfolio. DeMiguel et al. (2009) provide one of the most influential empirical challenges to sophisticated optimization by showing that many optimized strategies fail to consistently outperform the equally weighted $1/N$ portfolio out of sample. Kritzman et al. (2010) defend optimization against an overly literal interpretation of the $1/N$ result, but their response reinforces the same central issue: any optimization method must demonstrate that the economic value of its modeling structure exceeds the cost of estimation error. The

$1/N$ rule contains no expected-return forecast, no covariance inversion, and no complex optimization. Precisely because of this simplicity, it becomes a powerful benchmark for evaluating whether additional modeling structure creates genuine economic value or merely adds estimation noise. Recent evidence—paper by Feng (2026) published in *Financial Markets and Portfolio Management* extends this benchmarking logic to factor timing and volatility-managed strategies, showing that sophisticated timing and volatility-scaling rules must still be evaluated against the simple but demanding $1/N$ benchmark.

Robust portfolio optimization offered a formal response to this input-fragility problem by replacing point estimates with uncertainty sets and optimizing against adverse parameter realizations. In a generic robust mean-variance problem, the investor solves:

$$\max_w \min_{\mu \in U_{\mu}, \Sigma \in U_{\Sigma}} \left(w' \mu - \frac{\gamma}{2} w' \Sigma w \right).$$

Here U_{μ} and U_{Σ} represent plausible sets for expected returns and covariance matrices. Goldfarb and Iyengar (2003), Tütüncü and Koenig (2004), and Fabozzi et al. (2007) helped establish robust portfolio optimization as a bridge between financial economics and operations research. In this literature, uncertainty is no longer a distributional feature alone; it is also a model-risk and estimation-risk problem. The progression from Bayesian portfolio theory to Black-Litterman and robust optimization therefore represents one of the most important responses to the fragility of classical mean-variance optimization.

The sixth generation shifted the focus from return optimization to risk budgeting and dynamic risk scaling. Risk parity and risk budgeting represent an important practical evolution because they allocate risk contributions rather than capital weights. This is especially relevant when expected returns are noisy and traditional capital allocations conceal concentrated risk exposures. The risk contribution of asset i can be written as:

$$RC_i = w_i \frac{(\Sigma w)_i}{\sqrt{w' \Sigma w}}.$$

Risk parity seeks to equalize RC_i across assets or asset classes. Qian (2005) presents risk parity as a framework for true diversification, while Maillard et al. (2010) formalize equally weighted risk contribution portfolios. Asness et al. (2012) link risk parity to leverage aversion, Clarke et al. (2013) compare risk parity with maximum diversification and minimum variance, and López de Prado (2016) proposes hierarchical risk parity as a graph-based and clustering-oriented alternative to traditional optimization. Risk parity deserves separate treatment because it is not merely a technical weighting rule. It represents a major institutional shift from return forecasting to risk budgeting, and it has shaped large-scale asset allocation frameworks used by global macro funds, pension investors, and multi-asset managers.

A second branch of the same generation is volatility timing and volatility-managed portfolios. These models argue that investors should reduce exposure when volatility is high and increase exposure when volatility is low. This approach responds to a different limitation of static optimization: even if unconditional expected returns and covariances are difficult to estimate, con-

ditional volatility is persistent and can often be forecast more reliably. Fleming et al. (2001) show that volatility timing has economic value, Barroso and Santa-Clara (2015) demonstrate that volatility management can substantially reduce momentum crashes, Daniel and Moskowitz (2016) link momentum crashes to panic states and market rebounds, and Moreira and Muir (2017) show that volatility-managed factor portfolios can improve risk-adjusted performance across several factors. A representative volatility-managed return can be written as:

$$R_{t+1}^{VM} = \frac{c}{\hat{\sigma}_t^2} R_{t+1},$$

where $\hat{\sigma}_t^2$ is a forecast of conditional variance and c controls average exposure. The economic intuition is straightforward: if expected returns do not rise proportionally with volatility, reducing exposure during high-volatility periods can improve Sharpe ratios and reduce drawdowns.

The seventh generation developed regime-switching and state-dependent allocation models. While volatility management changes exposure as a function of conditional risk, regime-switching allocation allows the entire investment opportunity set to vary across macro-financial states. Expected returns, volatilities, correlations, factor premia, and tail dependence may all differ across recession, expansion, crisis, recovery, inflationary, or low-volatility regimes. Campbell and Viceira (1999) show that time-varying expected returns have important implications for long-horizon portfolio choice, Ang and Bekaert (2002) study international asset allocation under regime shifts, and Guidolin and Timmermann (2007) show that optimal allocations differ across crash, slow-growth, bull, and recovery regimes. In a state-dependent framework, portfolio weights can be expressed as:

$$w_t = \sum_{s=1}^S p_{t,s} w_s,$$

where $p_{t,s}$ is the probability of regime s at time t , and w_s is the allocation conditional on that regime. This generation shifts portfolio theory from static efficient frontiers and simple risk scaling to conditional allocation rules that depend on the prevailing market state.

The eighth generation connects predictive portfolio theory with parametric and machine-learning allocation. Before the recent machine-learning literature, Brandt et al. (2009) offered an important data-driven bridge by proposing parametric portfolio policies that map firm characteristics and conditioning variables directly into portfolio weights. Rather than first estimating expected returns and then optimizing, their approach models the portfolio policy itself. In simplified form, a characteristic-based portfolio rule can be written as:

$$w_{i,t} = \bar{w}_{i,t} + \frac{1}{N_t} \theta' x_{i,t},$$

where $x_{i,t}$ and θ denotes asset characteristics and how those characteristics tilt portfolio weights; $w_{i,t}$ and $\bar{w}_{i,t}$ are an asset's final weight and weight in a chosen benchmark portfolio, separately. The characteristics may include size, value, momentum, volatility, liquidity, or estimated factor exposures; they are not factor returns themselves. In the original Brandt et al. (2009)

formulation, the policy coefficients are estimated by maximizing the investor's sample average utility. More generally, the direct-policy framework can accommodate alternative portfolio objectives, constraints, transaction costs, and regularization. Parametric portfolio policies did not become a practitioner movement comparable to Black–Litterman or risk parity, but they are important because they anticipate machine-learning allocation: portfolio weights become functions of observable predictors. Parametric portfolio policy are therefore an important bridge from characteristic based investing to machine learning allocation.

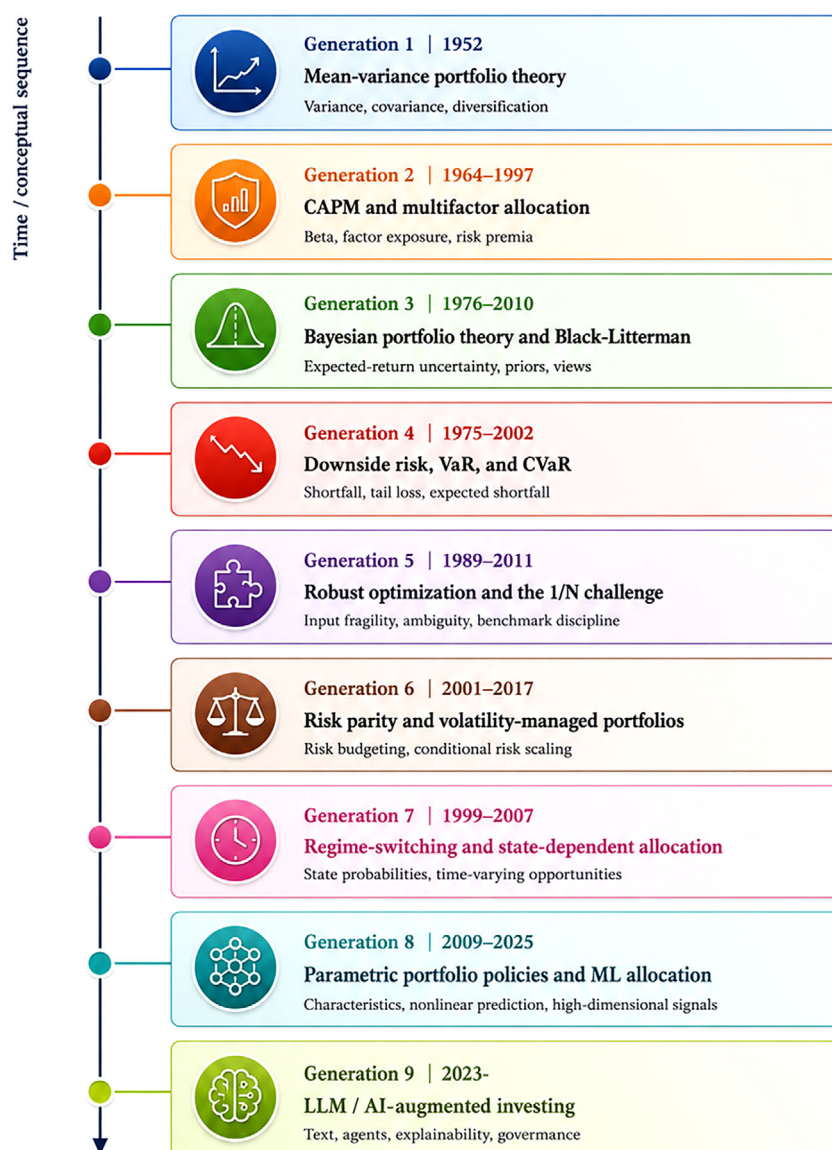
The most recent machine-learning generation expands this idea by using high-dimensional firm characteristics, macro variables, factor signals, text, news, and alternative data. Gu et al. (2020) show that machine-learning methods can capture nonlinear interactions and improve empirical asset-pricing performance. Chen et al. (2024) extend this agenda by using deep neural networks to estimate no-arbitrage asset-pricing models with flexible conditioning information. This generation changes the estimation layer of portfolio theory. Instead of treating expected returns, risks, regimes, and tail exposures as low-dimensional inputs, machine-learning methods attempt to learn them from high-dimensional and nonlinear information sets.

The ninth generation is shaped by large language models and agent-based investment systems. The emerging LLM literature goes beyond numerical prediction by using language models to process news, filings, reports, and market narratives. Lopez-Lira and Tang (2026) show that ChatGPT-based news scores contain predictive information for short-horizon stock returns, while FinGPT and FinMem illustrate how open-source financial language models and memory-based trading agents may support financial decision-making. This literature should be interpreted as an emerging frontier rather than as settled evidence comparable to classical portfolio-theory literature. Its promise lies in information processing, explanation, scenario monitoring, and workflow support; its risks include hallucination, prompt sensitivity, data leakage, overfitting, weak auditability, and unclear accountability.

The AI frontier does not replace portfolio theory. Instead, it adds a new information-processing and decision-support layer to the existing architecture. Machine learning can help forecast expected returns, volatility, tail risk, regimes, and factor exposures, while LLMs and agents can support text interpretation, scenario monitoring, explanation, and workflow automation. However, portfolio construction still requires the core elements developed by earlier generations: diversification, factor control, Bayesian updating, Black–Litterman views, downside-risk preferences, CVaR constraints, robust parameter treatment, risk-budgeting discipline, transaction-cost control, and governance. The next stage of portfolio theory is therefore likely to be AI-augmented rather than AI-determined.

Figure 1 summarizes this evolutionary structure. The figure is not intended to identify the first appearance of every technical method, but to provide a conceptual roadmap of the field. It shows how portfolio theory moved from mean–variance diversification to CAPM and multifactor allocation, Bayesian/Black–Litterman allocation, downside and tail-risk measures, robust

A vertical roadmap of nine generations of portfolio theory organized by the main form of risk, uncertainty, or information addressed.



Note: The figure organizes portfolio theory by conceptual turning points rather than by the first appearance of every technical method. Later generations do not replace earlier ones; they add new layers of uncertainty, implementation discipline, and information processing.

FIGURE 1 | Theoretical evolution from mean–variance to AI-augmented portfolio management. The figure organizes portfolio theory by conceptual turning points rather than by the first appearance of every technical method. Later generations do not replace earlier ones; they add new layers of uncertainty, implementation discipline, and information processing. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

optimization and the 1/N benchmark, risk parity and volatility management, regime-switching allocation, parametric portfolio policies and machine learning, and finally LLM/AI-augmented investing.

This article reviews the evolution of portfolio theory under risk and uncertainty from mean–variance optimization to AI-augmented portfolio management. It deliberately brings together six leading academic outlets—*Journal of Finance (JF)*, *Journal of Financial Economics (JFE)*, *Review of Financial Studies (RFS)*, *Review of Finance (RF)*, *Journal of Financial and Quantitative Analysis (JFQA)*, and *Management Science (MS)*—with four practitioner-oriented journals: *Financial Markets and Portfolio*

Management (FMPM), *Journal of Asset Management (JAM)*, *Journal of Portfolio Management (JPM)*, and *Financial Analysts Journal (FAJ)*. This dual focus is essential because portfolio theory has advanced through both formal academic innovation and implementation-oriented investment practice. The contribution of this article is to synthesize these literatures as a single evolutionary system. Rather than reviewing isolated techniques, the article shows how each generation of portfolio theory emerged in response to unresolved forms of uncertainty left by earlier models: mean–variance risk, systematic risk, expected-return uncertainty, downside and tail risk, parameter ambiguity, risk budgeting, conditional volatility, regime dependence, data-driven allocation, and AI-augmented decision-making.

TABLE 1 | Search protocol and screening design.

Component	Implementation
Time span	1952–2026
Main sources	Publisher/journal websites, DOI records, RePEc/EconPapers, JSTOR, SSRN, Google Scholar public pages, NBER, author PDFs
Top academic journals	JF, JFE, RFS, RF, JFQA, Management Science
Practical journals	FMPM, JAM, JPM, FAJ
Corpus structure	Core coded synthesis corpus plus supplementary methodological and implementation references
Main theoretical generations	Mean–variance; CAPM and multifactor allocation; Bayesian portfolio theory and Black–Litterman; downside risk, VaR, and CVaR; robust optimization and 1/N; risk parity and volatility management; regime-switching allocation; parametric portfolio policies and machine learning; LLM/AI-augmented investing
Coding fields	Generation, risk concept, method family, treatment of expected returns, outlet type, affiliation type, country/region
Expected-return coding	Direct sample estimate; equilibrium-implied return; Bayesian posterior; investor-view-adjusted return; robust uncertainty-set input; machine-learning forecast
Citation treatment	Public citation indicators used cautiously; no invented citation totals
Visual outputs	Nine generated figures based on the coded public-metadata corpus and thematic synthesis

2 | Methodology

This article uses a systematic literature review combined with public-metadata bibliometric analysis and selected thematic synthesis. The objective is not to reproduce a proprietary Web of Science or Scopus database, but to construct a transparent and replicable review corpus from publicly accessible records. The sources include publisher pages, journal websites, DOI pages, RePEc/EconPapers, JSTOR, SSRN, Google Scholar public pages, NBER working papers, and author-hosted PDFs. These records were used to verify titles, publication years, outlets, abstracts or abstract snippets, DOI information where available, author affiliations where visible, and public citation indicators where displayed. Citation counts are treated cautiously because Google Scholar, RePEc, SSRN, Wiley, Springer, Oxford, PM Research, and other platforms use different indexing and coverage rules. Therefore, citation information is used only as contextual evidence, not as a mechanically comparable ranking metric.

To make the review procedure transparent and replicable, Table 1 summarizes the search protocol, source types, journal groups, screening logic, and coding dimensions. The table clarifies that the corpus is not a convenience sample of well-known papers, but a structured public-metadata corpus designed to capture the main theoretical and practical generations in portfolio theory. Because the article's central purpose is to review how portfolio theory responds to different forms of uncertainty, the search and coding procedure was organized around conceptual generations rather than only journal ranking or citation counts.

Table 1 also explains why this survey separates top academic journals from practitioner-oriented journals. This distinction is important because portfolio theory has developed through both formal academic innovation and practical implementation. The top-journal group captures theoretical foundations and empirical breakthroughs, while FAJ, JPM, FMPM, and JAM capture how portfolio theory is translated into asset-management practice, risk

controls, institutional implementation, and investment governance. This separation is particularly important for topics such as Black–Litterman, covariance shrinkage, risk parity, hierarchical risk parity, and LLM-assisted investment management, where practitioner-oriented journals have played a central role in disseminating models that are widely used outside purely academic settings.

The search strategy combined conceptual and methodological terms across the nine generations. The main terms included: “portfolio selection,” “mean–variance,” “safety-first,” “CAPM,” “factor model,” “multifactor allocation,” “Bayesian portfolio theory,” “Bayesian asset allocation,” “Bayes–Stein estimation,” “Black–Litterman,” “Black–Litterman model,” “reverse optimization,” “equilibrium-implied returns,” “investor views,” “estimation error,” “1/N portfolio,” “downside risk,” “lower partial moment,” “semivariance,” “Value-at-Risk,” “Conditional Value-at-Risk,” “expected shortfall,” “robust portfolio optimization,” “uncertainty set,” “risk parity,” “risk budgeting,” “equal risk contribution,” “hierarchical risk parity,” “volatility-managed portfolios,” “volatility timing,” “regime-switching asset allocation,” “state-dependent allocation,” “parametric portfolio policies,” “characteristic-based portfolio allocation,” “data-driven portfolio construction,” “machine learning asset pricing,” “deep learning asset pricing,” “LLM investing,” and “financial trading agents.”

The inclusion criteria were: direct relevance to portfolio theory, asset allocation, risk measurement, optimization, portfolio implementation, or AI-augmented investment decision-making; verifiable public metadata; and contribution to a major theoretical or practical generation. Narrow trading-strategy papers without a clear portfolio-theory contribution were excluded. Purely technical optimization papers were also excluded unless they had direct relevance to investment allocation, risk measurement, or portfolio implementation. For the newest LLM and agent-investing literature, the inclusion criterion was intentionally stricter: papers were included only

when they connected language models or agent systems to return prediction, portfolio construction, investment management, or governance, rather than to general financial text analysis alone.

The final review corpus was coded along six dimensions. First, each study was assigned to one or more theoretical generations. Second, the core risk concept was identified, such as variance, beta, factor exposure, expected-return uncertainty, downside risk, tail risk, parameter ambiguity, risk contribution, conditional volatility, regime dependence, high-dimensional prediction, or AI governance. Third, the method family was coded, including mean–variance optimization, factor modeling, Bayesian allocation, Black–Litterman, lower partial moments, VaR/CVaR, robust optimization, shrinkage, 1/*N* benchmarking, risk parity, volatility scaling, regime switching, parametric portfolio policies, machine learning, deep learning, and LLM/agent systems. Fourth, the treatment of expected returns was coded separately because this is a central fault line in portfolio theory: expected returns may be treated as sample estimates, equilibrium-implied quantities, Bayesian posterior estimates, investor-view-adjusted quantities, robust uncertainty-set inputs, or machine-learning forecasts. Fifth, the outlet type was coded as top academic journal, practitioner-oriented journal, other academic outlet, working paper, book, or technical implementation reference. Sixth, visible author affiliations were coded as university/public research, industry/practitioner institution, or mixed academic–industry work.

The affiliation coding is important because portfolio theory is unusually connected to investment banks, asset managers, pension funds, risk-management institutions, and quantitative investment firms. For example, covariance shrinkage, Black–Litterman, risk parity, hierarchical risk parity, and LLM-assisted investment management all illustrate the interaction between academic modeling and practical portfolio construction. Ledoit and Wolf's covariance-shrinkage work links academic econometrics with investment implementation, while Satchell and Scowcroft's JAM discussion of Black–Litterman illustrates how Bayesian portfolio construction entered asset-management practice. This hybrid structure is one reason why the review deliberately includes FAJ, JPM, FMPM, and JAM alongside JF, JFE, RFS, RF, JFQA, and Management Science.

Figure 2 presents the screening flow used to construct the final review corpus. It reports how the initial pool of public records was reduced through duplicate removal, relevance screening, full-metadata verification, and thematic coding. This figure is included to make clear that the bibliometric and thematic analysis is based on a defined review corpus rather than on an informal selection of influential papers.

3 | Bibliometric Analysis

3.1 | The Historical Structure of the Field

The bibliometric structure of portfolio theory is best understood as a sequence of intellectual waves rather than a smooth accumulation of papers. These waves are not mutually exclusive, and later generations do not simply replace earlier ones. Instead, each gen-

eration responds to a different unresolved form of uncertainty left by previous models. The first wave is the classical portfolio-theory wave. Roy (1952) and Markowitz (1952) appeared in the same year but represent two distinct risk languages: safety-first shortfall and mean–variance diversification. Markowitz became dominant because variance and covariance are mathematically tractable, while Roy's safety-first idea later reappeared in downside-risk and target-based portfolio choice. Markowitz's article remains the central citation anchor of the corpus, while his later book and Nobel lecture consolidated the framework as the foundation of modern portfolio theory (Markowitz 1959, 1991).

The second wave is the equilibrium-pricing and multifactor-allocation wave. Sharpe (1964), Lintner (1965), Mossin (1966), and Black (1972) transformed the Markowitz framework into a market-pricing theory. The CAPM made systematic market beta the relevant risk measure, while Ross's arbitrage pricing theory later generalized pricing to multiple factors. This wave is highly concentrated in top academic journals and economics outlets. Its main intellectual movement is from individual portfolio choice to equilibrium restrictions on expected returns. Fama and French (1992, 1993) and Carhart (1997) then changed portfolio theory by making factor exposure central to asset management. Jegadeesh and Titman (1993), Asness et al. (2013), Frazzini and Pedersen (2014), and Fama and French (2015) extend this factor-based view. The investor is no longer only choosing securities; the investor is choosing exposures to systematic premia.

The third wave is Bayesian portfolio theory and Black–Litterman allocation. This wave responds to one of the most important practical weaknesses of the Markowitz framework: the difficulty of estimating expected returns. Expected returns are usually harder to estimate than covariances, and small errors in expected returns can produce extreme portfolio weights. Klein and Bawa (1976) and Jorion (1986) address estimation risk through Bayesian and Bayes–Stein approaches, while Pástor (2000), Pástor and Stambaugh (2000), and Avramov and Zhou (2010) develop broader Bayesian perspectives on portfolio choice and model uncertainty. Black and Litterman (1992) provide the most influential practitioner-facing version of this logic by combining equilibrium-implied returns with investor views. Satchell and Scowcroft (2000), He and Litterman (2002), Idzorek (2004), and Meucci (2010) further explain and extend the model for implementation. This wave is important because it changes expected returns from unstable direct estimates into equilibrium-implied, Bayesian, and view-adjusted quantities.

The fourth wave is downside risk, VaR, coherent risk, and CVaR. Bawa (1975), Fishburn (1977), Hogan and Warren (1974), Harlow and Rao (1989), Harlow (1991), and Sortino and van der Meer (1991) shift attention from variance to below-target risk. This literature responds to the symmetry problem of variance: investors usually care more about losses, shortfalls, and drawdowns than about upside variation. The VaR and coherent-risk literature then shifts the focus from below-target shortfall to institutional tail-risk measurement. Artzner et al. (1999) provide the axiomatic critique of incoherent risk measures, while Rockafellar and Uryasev (2000, 2002) provide the optimization engine for CVaR. Acerbi and Tasche (2002) further clarify expected shortfall as a coherent alternative to VaR.

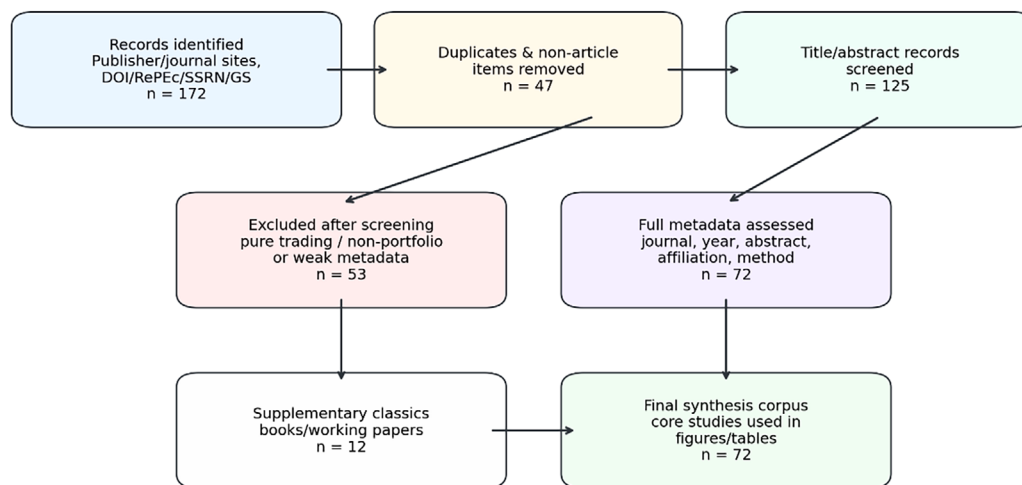


FIGURE 2 | Search and screening flow for the public-metadata corpus. The figure reports the screening process for the curated review corpus used in the bibliometric and thematic analysis. It is a transparent review corpus, not a claim about the full universe of portfolio-theory publications. As Figure 2 shows, the final coded corpus balances historical classics, top-journal theoretical contributions, practitioner-oriented implementation papers, and emerging AI/LLM studies. This structure is particularly suitable for a survey of portfolio theory because the field has always advanced through interaction between academic finance, operations research, investment management, and institutional practice. Because this survey relies on publicly accessible metadata rather than a proprietary Web of Science or Scopus export, the bibliometric results should be interpreted as structured evidence from a transparent review corpus rather than as full-population estimates of the entire portfolio-theory literature. The purpose of the bibliometric analysis is therefore not to rank papers mechanically by citation counts, but to identify intellectual communities, outlet structures, methodological clusters, and the institutional channels through which portfolio theory has evolved. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/joes.70148)]

The fifth wave is robust optimization and the $1/N$ challenge. Michaud (1989), Best and Grauer (1991), Chopra and Ziemba (1993), Jagannathan and Ma (2003), Ledoit and Wolf (2004), Kan and Zhou (2007), DeMiguel et al. (2009), Kritzman et al. (2010), and Tu and Zhou (2011) all respond to the same empirical difficulty: optimized portfolios are highly sensitive to uncertain inputs. DeMiguel et al. (2009) made the $1/N$ portfolio a central benchmark by showing that many optimized models do not consistently outperform naïve diversification out of sample. Robust optimization, especially Goldfarb and Iyengar (2003), Tütüncü and Koenig (2004), and Fabozzi et al. (2007), converts this difficulty into a formal allocation problem under uncertainty sets. This wave therefore treats uncertainty as input fragility, parameter ambiguity, and model risk.

The sixth wave is risk parity, risk budgeting, and volatility-managed portfolios. This wave is more practitioner-oriented because it directly addresses weight concentration, instability, leverage, drawdowns, and implementability. Risk parity shifts the allocation object from capital weights to risk contributions. Qian (2005), Maillard et al. (2010), Asness et al. (2012), Clarke et al. (2013), and López de Prado (2016) show how risk-budgeting ideas can reduce reliance on expected-return estimates and produce more diversified risk exposure. Volatility management is the dynamic counterpart of this risk-control logic. Fleming et al. (2001), Barroso and Santa-Clara (2015), Daniel and Moskowitz (2016), and Moreira and Muir (2017) show that risk exposure should vary over time when conditional volatility is predictable.

The seventh wave is regime-switching and state-dependent allocation. While volatility management scales exposure with conditional risk, regime-switching allocation allows the investment opportunity set itself to change across states. Campbell and Viceira (1999), Ang and Bekaert (2002), and Guidolin and

Timmermann (2007) show that expected returns, volatilities, correlations, and optimal allocations can differ across long-horizon, international, and macro-financial regimes. This wave shifts portfolio theory from static efficient frontiers and simple risk scaling to conditional allocation rules based on state probabilities.

The eighth wave is parametric portfolio policies and machine-learning allocation. Brandt et al. (2009) develop parametric portfolio policies that map firm characteristics and conditioning variables directly into portfolio weights. This approach does not constitute a full independent era comparable to Black–Litterman, CVaR, robust optimization, or risk parity, but it provides an important bridge from predictive portfolio theory to machine-learning allocation. Kelly et al. (2019), Freyberger et al. (2020), Feng et al. (2020), Gu et al. (2020), Bryzgalova et al. (2025), and Chen et al. (2024) extend this data-driven logic by using high-dimensional characteristics, nonlinear functions, latent factors, and deep learning to estimate risk premia and portfolio-relevant inputs.

The ninth wave is LLM- and AI-augmented investing. This literature is still emerging and should not be treated as settled evidence comparable to the classical portfolio-theory literature. Nevertheless, Lopez-Lira and Tang (2026), FinGPT, FinMem, and Kong et al. (2024) show that large language models and agentic systems may affect portfolio management through text interpretation, signal extraction, scenario monitoring, investment explanations, and governance workflows. The historical development of the field is summarized in Figure 1. Rather than treating portfolio theory as a set of disconnected techniques, the figure organizes the literature into nine successive waves: mean–variance theory; CAPM and multifactor allocation; Bayesian portfolio theory and Black–Litterman; downside risk, VaR, and CVaR; robust optimization and $1/N$; risk parity and volatility management;

regime-switching allocation; parametric portfolio policies and machine-learning allocation; and LLM/AI-augmented investing.

Figure 1 highlights the cumulative nature of portfolio theory. Later approaches do not simply replace Markowitz's framework; they add new layers of uncertainty. CAPM and factor models address systematic risk; Bayesian portfolio theory and Black–Litterman address expected-return uncertainty; downside-risk models address the symmetry problem of variance; CVaR addresses the severity of tail losses; robust optimization addresses input uncertainty; risk parity addresses risk concentration; volatility-managed portfolios address conditional risk; regime-switching models address state dependence; and AI-based methods address high-dimensional information and decision support.

3.2 | Outlet Structure: Top Journals and Practical Journals

A major bibliometric finding is that portfolio theory has a dual outlet structure. Top journals produce the formal foundations and empirical breakthroughs, while practical journals translate them into implementable investment methods. In the selected corpus, the top-journal cluster is strongest in CAPM and multifactor theory, robust constraints, volatility management, and ML/deep-learning asset pricing. The practical-journal cluster is strongest in Bayesian implementation, Black–Litterman allocation, downside-risk performance measurement, estimation-risk correction, covariance shrinkage, risk parity, hierarchical risk parity, and institutional portfolio construction.

This is visible in the role of FAJ, JPM, and JAM. Michaud's FAJ article crystallized the practitioner critique of mean-variance optimization, while Black and Litterman (1992) created a Bayesian portfolio framework widely adopted in asset management. Satchell and Scowcroft's JAM article explicitly presents Black–Litterman as a Bayesian portfolio construction method used in the asset-management industry. This makes Black–Litterman more than an implementation detail. It is a central bridge between academic portfolio theory and asset-management practice because it transforms expected returns from unstable direct estimates into equilibrium-implied, view-adjusted posterior quantities. JPM's Ledoit and Wolf article addressed covariance estimation error, and López de Prado's JPM article introduced hierarchical risk parity as a response to instability, concentration, and out-of-sample underperformance.

FMPM and JAM play a different but important role. FMPM is particularly valuable for survey, implementation, and European portfolio-management research. Koumou's FMPM review explicitly organizes diversification around the law of large numbers, correlation, CAPM, and risk contribution/risk parity principles. Ben Salah et al. (2018) extend mean-variance and mean-semivariance portfolio selection using a multivariate nonparametric approach in FMPM, directly linking downside risk with modern estimation methods. JAM contributes applied asset-management perspectives, including Black–Litterman, uncertain-world portfolio optimization, and factor allocation. Bessler, Taushanov, and Wolff's JAM article compares factor and

sector allocation strategies, reflecting the practitioner shift from country allocation to factor and style allocation.

Figure 3 compares the dominant theoretical generations across outlet types. The figure separates top academic journals from practitioner-oriented journals and other academic outlets in order to show where different parts of portfolio theory have been developed and disseminated.

Figure 3 shows that classical theory, CAPM/factor allocation, volatility management, and ML/AI asset pricing are more concentrated in top academic journals, whereas Bayesian implementation, Black–Litterman, robust implementation, risk budgeting, and practitioner-facing portfolio construction are more visible in FAJ, JPM, FMPM, and JAM. This confirms that the evolution of portfolio theory cannot be understood solely through top-journal citations; practitioner outlets play a central role in turning theory into implementable asset-management frameworks.

Table 2 provides a more detailed interpretation of this outlet structure. It identifies the distinctive role of each journal group in the evolution of portfolio theory, distinguishing theoretical foundations, empirical asset pricing, Bayesian and Black–Litterman implementation, robust allocation, risk parity, and applied portfolio-management research. The table shows that FMPM and JAM should not be treated as marginal outlets in this review. FMPM contributes to the survey and implementation literature on diversification, semivariance, behavioral finance, and portfolio construction, while JAM provides an applied asset-management channel for Black–Litterman models, uncertain-world optimization, and factor-allocation strategies. Together with FAJ and JPM, these journals form the practical bridge between academic theory and investment practices.

Figure 4 further decomposes the academic-practical distinction by topic. While Figure 3 compares theoretical generations, Figure 4 focuses on the relative emphasis placed on different research themes by top journals and practitioner-oriented outlets. The figure reveals a clear division of labor. Top journals dominate factor allocation, volatility management, and ML/AI research, whereas practical journals are especially important for Bayesian implementation, Black–Litterman, downside risk, estimation error, risk parity, and implementable robust allocation. This pattern supports the article's broader argument that portfolio theory evolves through both formal modeling and practical adaptation.

Before turning to author affiliations and institutional origins, Figure 5 maps the keyword co-occurrence structure of the selected corpus. The network is constructed from recurring terms in titles, abstracts, and publisher summaries. It is designed to show the conceptual communities that organize the field rather than to provide a full automated text-mining exercise. Figure 5 identifies broad communities around classical mean-variance theory, factor allocation, Bayesian/Black–Litterman allocation, downside and tail risk, robust estimation, dynamic allocation, and AI-based portfolio management. The network illustrates that these communities are connected rather than isolated. For example, Black–Litterman links equilibrium asset

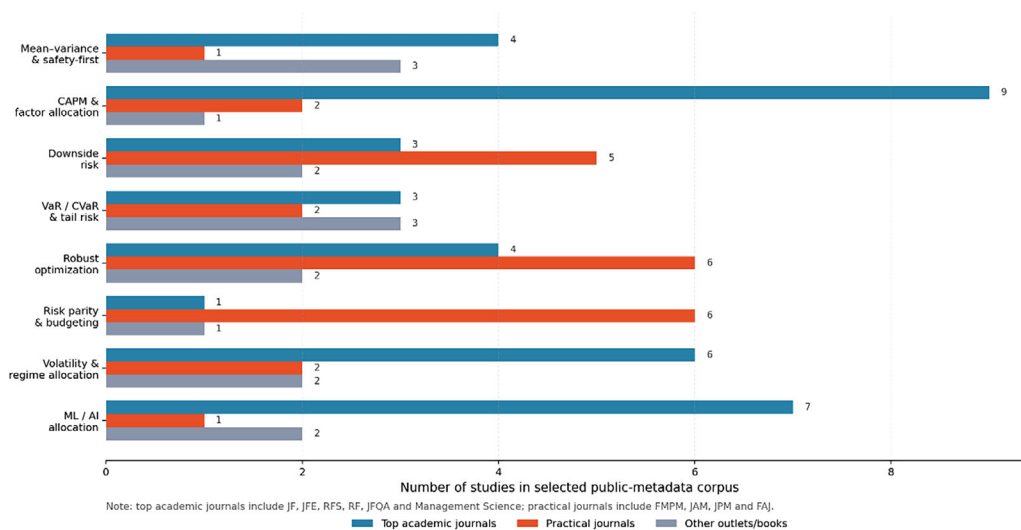


FIGURE 3 | Dominant generations by outlet type in the selected corpus. Counts refer to the coded public-metadata corpus. The practical-journal cluster is particularly important for Bayesian/Black–Litterman implementation, robust/estimation-risk methods, risk parity, and hierarchical risk parity. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 2 | Outlet roles in the evolution of portfolio theory.

Outlet group	Main contribution	Representative studies
JF, JFE, RFS, RF, JFQA, Management Science	Formal theory, equilibrium asset pricing, empirical asset pricing, robust constraints, volatility management, ML/deep-learning frontier	Markowitz; Sharpe; Fama–French; Jagannathan–Ma; DeMiguel et al.; Moreira–Muir; Gu–Kelly–Xiu; Chen–Pelger–Zhu
FAJ	Practitioner theory, institutional implementation, Bayesian allocation, Black–Litterman, downside risk, risk parity	Michaud; Black–Litterman; Harlow; Asness–Frazzini–Pedersen
JPM	Covariance shrinkage, input-error discipline, risk parity, HRP, LLM investment management, investment implementation	Chopra–Ziemba; Ledoit–Wolf; Maillard–Roncalli–Teiletche; López de Prado; Kong et al.
FMPM	Diversification surveys, semivariance, behavioral finance, robust/portfolio applications	Koumou; Ben Salah et al.; De Giorgi–Hens
JAM	Fund-management practice, Black–Litterman implementation, uncertain-world optimization, factor allocation	Satchell–Scowcroft; de Jong; Bessler et al.

pricing, Bayesian updating, and practitioner implementation; robust optimization links classical mean–variance analysis with parameter uncertainty; and ML/AI connects factor allocation, regime detection, text-based investment signals, and explainable decision systems. This supports the survey’s central claim that portfolio theory develops through cumulative extensions of uncertainty.

3.3 | Author Affiliations, Countries, and Academic–Industry Interaction

Portfolio theory is unusually hybrid in terms of authorship. Markowitz’s work was developed at the Cowles Commission, an academic research environment devoted to formal economics. But later generations increasingly include investment banks,

asset managers, quant firms, and professional risk institutions. This is not incidental. Portfolio theory is a field where models are judged not only by internal mathematical coherence but also by whether they survive estimation error, turnover, transaction costs, leverage constraints, and investor governance.

The Bayesian, Black–Litterman, and estimation-risk literatures illustrate this hybrid structure particularly clearly. Black–Litterman originated in an investment-bank setting and later became a central practitioner-facing framework for combining equilibrium-implied returns with investor views. Satchell and Scowcroft’s JAM treatment of Black–Litterman connects Cambridge University and UBS Warburg affiliations, illustrating the academic–industry channel through which Bayesian allocation entered asset-management practice. The estimation-risk literature has a similar mixed structure. Ledoit and Wolf’s covariance-

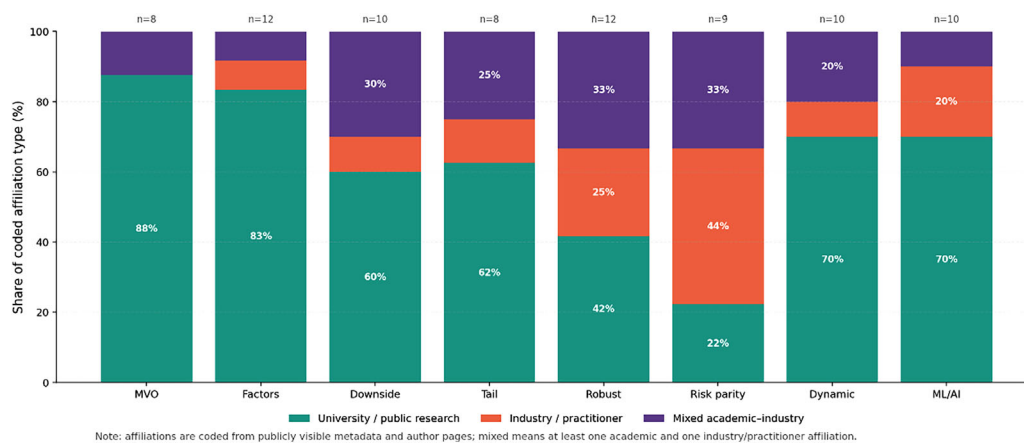


FIGURE 6 | Author-affiliation mix by theoretical generation. This figure codes author-paper affiliations as university/public research, asset managers/banks/vendors, or mixed academic-industry. It highlights the strong industry role in Black-Litterman, robust estimation, risk parity, and AI/agentive investing. [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

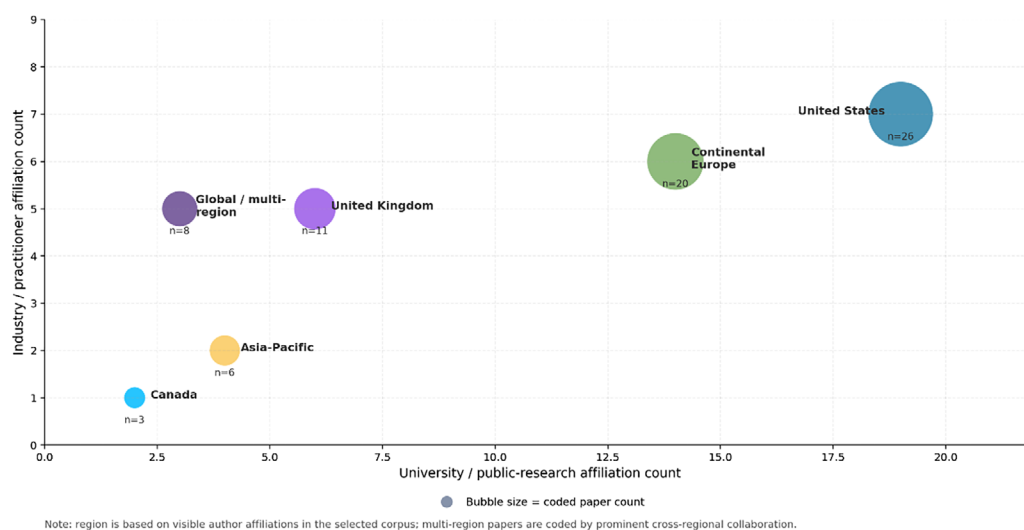


FIGURE 7 | Country/region and sector profile of contributors. The selected corpus is dominated by the United States and Western Europe, but the practical-journal channel gives a larger role to industry-linked work in the United Kingdom, Switzerland, France, and global asset-management organizations. [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

agent communities. The NBER version of Kelly, Pruitt, and Su's paper acknowledges AQR Capital Management in the research environment.

Figure 6 reports the affiliation mix by theoretical generation. This figure is important because portfolio theory is not a purely academic literature. Several major advances were developed or popularized by authors connected to banks, asset managers, quant firms, risk-management institutions, or mixed academic-industry collaborations. Figure 6 shows that university-based research dominates the classical, factor, and ML/AI generations, but industry and mixed affiliations become more visible in Bayesian/Black-Litterman allocation, robust optimization, risk parity, and LLM/agent investing. This pattern is economically intuitive: the closer a topic is to implementation, portfolio constraints, model governance, and trading workflow, the stronger the role of practitioner institutions becomes.

Figure 7 extends the affiliation analysis by showing the country/region and sector profile of contributors. The purpose is not

to rank countries, but to identify where the academic-industry interaction in portfolio theory is most visible. Figure 7 indicates that the United States and Western Europe dominate the selected corpus, with important contributions from the United Kingdom, Switzerland, France, and other European financial centers. This geographical structure reflects the institutional ecology of the field: major universities, central financial markets, global asset managers, investment banks, and quantitative research groups are all involved in developing portfolio theory. The figure also helps explain why practical journals such as FAJ, JPM, FMPM, and JAM are important channels for this literature.

3.4 | Citation Anchors and Intellectual Communities

The citation structure of the field is not just a ranking of famous papers. It is a map of conceptual dependency. Markowitz is cited because later work either extends mean-variance analysis or responds to its weaknesses. Sharpe is cited because CAPM

redefines risk as beta. Fama and French are cited because factor models become the language of empirical portfolio allocation. Jorion, Black–Litterman, Satchell–Scowcroft, Idzorek, Meucci, Pástor, Pástor–Stambaugh, and Avramov–Zhou are cited because Bayesian portfolio theory and equilibrium-implied returns address expected-return uncertainty. Artzner et al. and Rockafellar–Uryasev are cited because tail risk becomes formal and optimizable. Michaud, Chopra–Ziemba, Jagannathan–Ma, Ledoit–Wolf, Kan–Zhou, DeMiguel et al., and Kritzman et al. are cited because they address input fragility and the discipline imposed by $1/N$. Maillard–Roncalli–Teiletche, Asness–Frazzini–Pedersen, Qian, Clarke–de Silva–Thorley, and López de Prado are cited because risk budgeting changes the allocation object from capital weights to risk contributions. Moreira–Muir and Barroso–Santa-Clara are cited because they show that risk exposure should vary over time. Brandt–Santa-Clara–Valkanov are cited because parametric portfolio policies provide a bridge to data-driven allocation. Gu–Kelly–Xiu and Chen–Pelger–Zhu are cited because modern portfolio theory increasingly relies on nonlinear prediction.

Table 3 organizes the main citation-anchor communities identified in the review. Instead of ranking papers only by citation counts, the table classifies them by conceptual role: classical theory, equilibrium and factor models, Bayesian and Black–Litterman allocation, downside risk, coherent and tail risk, robust estimation and $1/N$, risk budgeting, dynamic allocation, parametric/ML allocation, and LLM/AI. This classification is more useful for a survey article than a simple citation ranking. For example, Markowitz (1952), Black and Litterman (1992), Rockafellar and Uryasev (2000), Ledoit and Wolf (2004), Moreira and Muir (2017), Brandt et al. (2009), and Gu et al. (2020) belong to different citation communities because they redefine different dimensions of uncertainty. The table therefore helps readers understand how the field is structured intellectually, not merely which papers are famous.

3.5 | Methodological Mapping

The methodological evolution of portfolio theory can be summarized by the movement from closed-form optimization to data-driven decision systems. Mean–variance optimization requires expected returns and covariances. CAPM and multifactor allocation require systematic exposures. Bayesian portfolio theory and Black–Litterman require priors, equilibrium-implied returns, investor views, and confidence levels. Downside-risk optimization requires target-dependent shortfall moments. CVaR requires loss scenarios and tail expectations. Robust optimization requires uncertainty sets. The $1/N$ benchmark requires no estimated expected returns and therefore disciplines the entire optimization literature. Risk parity requires marginal risk contributions. Volatility-managed portfolios require conditional volatility forecasts. Regime-switching allocation requires latent state probabilities. Parametric portfolio policies require a benchmark portfolio, standardized asset characteristics, and a low dimensional vector of policy coefficients that maps those characteristics into weights. Machine learning requires high-dimensional features, nonlinear function approximation, and out-of-sample validation. LLM/agent investing requires text

ingestion, retrieval, reasoning, memory, action constraints, and governance.

Figure 8 summarizes the relationship between method families and uncertainty dimensions. The heatmap does not treat methods as substitutes. Instead, it shows which uncertainty problems each method is designed to address.

Figure 8 shows that no single method covers all dimensions of uncertainty. Mean–variance optimization is strong on covariance-based risk but weak on tail risk and parameter ambiguity. CAPM and factor models address systematic exposure but do not solve estimation error. Bayesian/Black–Litterman allocation directly addresses expected-return uncertainty by replacing raw expected-return estimates with equilibrium-implied and view-adjusted posterior quantities. CVaR is strong on tail severity but depends on scenarios. Robust optimization addresses ambiguity but may be conservative. Risk parity reduces dependence on expected returns but still depends on covariance stability. Volatility management addresses conditional risk but not broader state dependence. Regime-switching allocation addresses state dependence but introduces real-time classification uncertainty. Parametric portfolio policies and ML/AI are powerful for high-dimensional prediction but require robust risk filters, explainability, transaction-cost discipline, and governance. This interpretation motivates the integrative research agenda developed in Section 4.

Table 4 provides a formula roadmap for the survey. Because portfolio theory is mathematically cumulative, the table helps readers see how each generation modifies the object of optimization, the risk measure, the expected-return input, or the information set. This table also clarifies why the article should be read as a theory-evolution survey rather than as a technical review of one optimization method. The formulas move from variance to beta, factors, Bayesian posterior returns, lower partial moments, VaR, CVaR, robust uncertainty sets, risk contributions, volatility scaling, regime probabilities, parametric weight rules, and AI-based allocation rules. This progression is the mathematical backbone of the survey.

Overall, the bibliometric analysis shows that portfolio theory has evolved through a set of connected intellectual communities rather than through isolated methodological innovations. Figures 2–8 and Tables 2–4 jointly show three patterns. First, the field has moved from variance-based diversification to increasingly rich concepts of uncertainty, including systematic risk, expected-return uncertainty, Bayesian posterior beliefs, downside risk, tail loss, parameter ambiguity, conditional volatility, regime dependence, high-dimensional prediction, and AI governance. Second, the literature is institutionally hybrid: top academic journals provide theoretical and empirical foundations, while FAJ, JPM, FMPM, and JAM translate these ideas into implementable portfolio-management frameworks. Third, no single method dominates all dimensions of uncertainty. These findings motivate the selected thematic review in Section 4, which explains how each generation of portfolio theory modifies the previous one and why the current frontier lies in AI-augmented, robust, tail-aware, regime-sensitive, and institutionally governed portfolio systems.

TABLE 3 | Citation-anchor communities and their conceptual role.

Community	Anchor papers	Central concept	Main unresolved problem
Classical portfolio theory	Markowitz; Roy; Tobin	Diversification and safety-first risk	Symmetric variance; static setting
Equilibrium/factor models	Sharpe; Lintner; Mossin; Ross; Fama–French; Carhart	Systematic risk and factor exposure	Factor instability; factor zoo
Bayesian and Black–Litterman allocation	Klein–Bawa; Jorion; Black–Litterman; Satchell–Scowcroft; He–Litterman; Idzorek; Meucci; Pástor; Pástor–Stambaugh; Avramov–Zhou	Posterior expected returns, reverse optimization, equilibrium-implied returns, investor views	Prior sensitivity; view confidence; implementation discipline
Downside-risk theory	Bawa; Fishburn; Harlow; Sortino	Shortfall and lower partial moments	Target choice; tail severity
Coherent/tail risk	Artzner et al.; Rockafellar–Uryasev; Acerbi–Tasche	VaR, CVaR, expected shortfall	Scenario dependence; rare-event data
Robust estimation and 1/N	Michaud; Best–Grauer; Chopra–Ziembra; Jagannathan–Ma; Ledoit–Wolf; Kan–Zhou; DeMiguel et al.; Kritzman et al.; Tu–Zhou; Feng	Input uncertainty, shrinkage, uncertainty sets, benchmark discipline	Conservatism; calibration; whether sophistication beats simple diversification
Risk budgeting	Qian; Maillard–Roncalli–Teïletche; Asness–Frazzini–Pedersen; Clarke–de Silva–Thorley; López de Prado	Equal risk contribution, risk parity, HRP	Leverage, correlation instability, crowding
Dynamic allocation	Campbell–Viceira; Ang–Bekaert; Guidolin–Timmermann; Fleming et al.; Barroso–Santa-Clara; Daniel–Moskowitz; Moreira–Muir	Conditional variance, volatility scaling, regimes	Real-time state identification; timing instability
Parametric and ML allocation	Brandt–Santa-Clara–Valkanov; Kelly–Pruitt–Su; Freyberger–Neuhierl–Weber; Feng–Giglio–Xiu; Gu–Kelly–Xiu; Chen–Pelger–Zhu	Characteristics, nonlinear prediction, learned allocation rules	Overfitting, transaction costs, interpretability
LLM/AI	Lopez-Lira–Tang; FinGPT; FinMem; Kong et al.	Text, narratives, agents, decision support	Hallucination, leakage, explainability, governance

4 | Selected Thematic Review

4.1 | Markowitz and the Enduring Grammar of Portfolio Choice

The Markowitz framework remains the grammar of portfolio theory because it established three ideas that still survive in modern portfolio management: expected return, risk, and portfolio constraints. Even when later models reject variance, use Bayesian priors, impose robust uncertainty sets, allocate by risk contribution, or rely on machine learning, they still solve a version of the same economic problem: how to transform

uncertain future payoffs into feasible portfolio weights. The classical mean–variance problem can be written as:

$$\min_w w' \Sigma w$$

subject to:

$$w' \mu \geq \mu^*, 1' w = 1$$

This formulation shows why Markowitz (1952) remains foundational. The efficient frontier is not only a diagram; it is a theory of disciplined trade-offs. Portfolio choice becomes a constrained

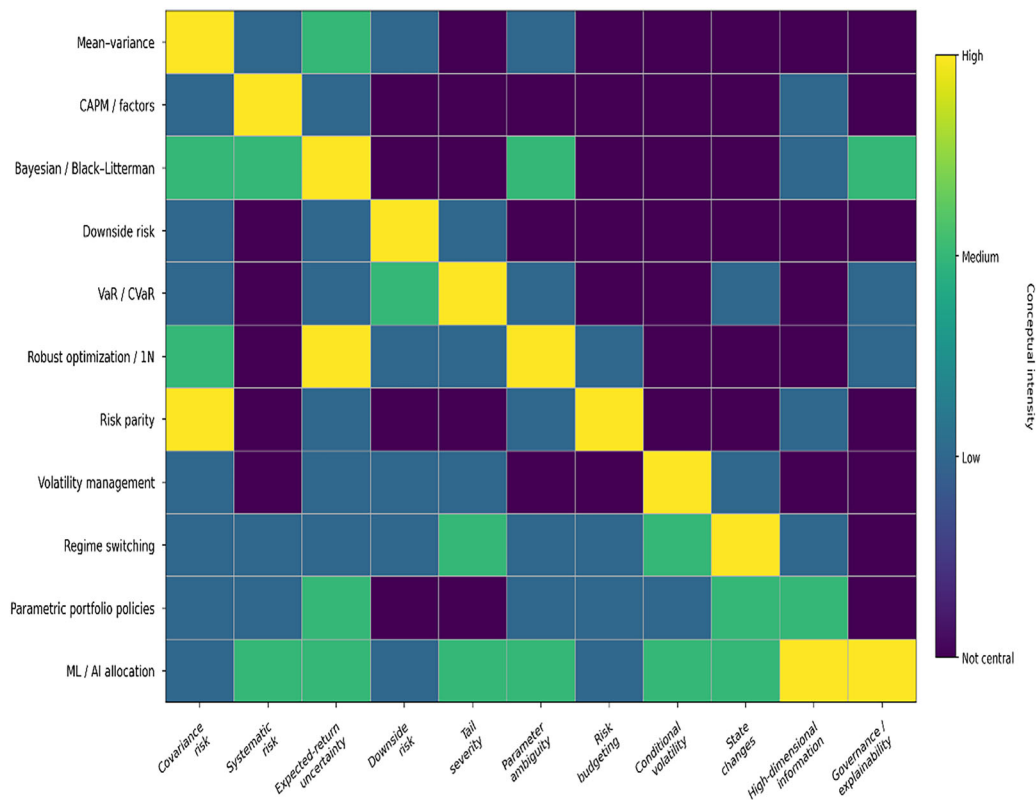


FIGURE 8 | Method-uncertainty matrix. The heatmap summarizes the conceptual intensity with which each method family addresses different uncertainty dimensions. Scores are hand-coded from the selected corpus and used only for interpretive visualization. The revised matrix separates Bayesian/Black-Litterman allocation, robust/1N benchmarking, parametric portfolio policies, and ML/AI allocation. [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 4 | Formula roadmap across generations.

Generation	Core formula	Economic interpretation
Mean-variance	$\max_w (w' \mu - \frac{\gamma}{2} w' \Sigma w)$	Trade expected return against variance
CAPM	$E(R_i - R_f) = \beta_i E(R_M - R_f)$	Only systematic market risk is priced
Factor allocation	$R_p - R_f = \alpha_p + \beta_p' f + \varepsilon_p$	Portfolios are bundles of factor exposures
Bayesian/Black-Litterman	$\mu_{BL} = [(\tau \Sigma)^{-1} + P' \Omega^{-1} P]^{-1} [(\tau \Sigma)^{-1} \pi + P' \Omega^{-1} Q]$	Expected returns become equilibrium-implied, view-adjusted posterior quantities
Downside risk	$LPM_n(\tau) = E[\max(\tau - R_p, 0)^n]$	Risk is below-target shortfall
VaR	$VaR_\alpha(L) = \inf\{l : P(L \leq l) \geq \alpha\}$	Risk is quantile loss
CVaR	$CVaR_\alpha = \min_\eta [\eta + \frac{E(L-\eta)^+}{1-\alpha}]$	Risk is expected loss in the tail
Robust optimization	$\max_w \min_{\theta \in \mathcal{U}} U(w, \theta)$	Inputs are uncertain within a plausible set
1/N benchmark	$w_i = \frac{1}{N}$	Simple diversification disciplines complex optimization
Risk parity	$R C_i = \frac{w_i(\Sigma w)_i}{\sqrt{w' \Sigma w}}$, equalized across assets	Allocate risk, not capital
Volatility management	$R_{t+1}^{VM} = \frac{c}{\hat{\sigma}_t^2} R_{t+1}$	Reduce exposure when volatility is high
Regime allocation	$w_t = \sum_s p_{t,s} w_s$	Allocate conditional on state probabilities
Parametric portfolio policies	$w_{i,t} = w_{i,t} + N_t^{-1} \theta' x_{i,t}$	Map characteristics directly into portfolio weights
ML/AI allocation	$w_t = g_\theta(X_t)$ subject to constraints	Learn nonlinear allocation rules from high-dimensional data

decision problem in which expected returns, covariance, and feasibility jointly determine investment allocation. The weakness of the framework is equally important. The optimizer depends heavily on μ and Σ , treats upside and downside volatility symmetrically, and assumes a stable single-period investment environment. These weaknesses motivated much of the later literature, from Bayesian portfolio theory and Black–Litterman to Michaud’s FAJ critique, Ledoit–Wolf shrinkage, robust optimization, and DeMiguel et al.’s $1/N$ benchmark.

The key point is not that Markowitz is obsolete. Rather, later portfolio theory repeatedly modifies one of four elements of the Markowitz problem: the risk measure, the expected-return input, the information set, or the constraints. Downside-risk theory changes the risk measure. Black–Litterman changes how expected returns are formed. Robust optimization changes the treatment of input uncertainty. Risk parity changes the allocation object from capital weights to risk contributions. Volatility management and regime switching make portfolio weights conditional on market states. Machine learning and AI change the estimation and information-processing layer. The rest of this section reviews these extensions as cumulative responses to unresolved forms of uncertainty.

4.2 | CAPM and Multifactor Allocation: From Securities to Systematic Exposures

The first major extension of the Markowitz framework was equilibrium asset pricing. CAPM transformed portfolio theory from a theory of efficient diversification into a theory of priced systematic risk. In the CAPM, the expected excess return of asset i is determined by its exposure to the market portfolio:

$$E(R_i - R_f) = \beta_i E(R_M - R_f).$$

Sharpe (1964), Lintner (1965), Mossin (1966), and Black (1972) therefore changed the central object of risk from total variance to market beta. Ross (1976) further generalized this logic through arbitrage pricing theory, allowing multiple sources of systematic risk. The conceptual contribution is important for portfolio management because it shifts attention from individual asset selection to systematic exposure management. A portfolio is no longer simply a collection of securities; it is a bundle of risk exposures.

The empirical factor literature extended this logic. Fama and French (1993) introduced common stock and bond factors, while Carhart (1997) incorporated momentum into performance evaluation. Jegadeesh and Titman (1993) documented the momentum effect, and Asness et al. (2013) show that value and momentum are pervasive across markets and asset classes. In factor form, portfolio returns can be written as:

$$R_p - R_f = \alpha_p + \beta_p' f + \varepsilon_p,$$

where f denotes a vector of factor returns and β_p denotes the portfolio’s factor exposures. This representation made factor allocation central to asset management. The investor is not only choosing securities; the investor is choosing exposures to market,

size, value, momentum, profitability, investment, low-risk, and other systematic premia.

The limitation is factor instability. Factors may be time-varying, crowded, regime-dependent, expensive to trade, or vulnerable to data mining. The factor zoo problem also raises the question of which factors are genuine economic risk premia, which are behavioral anomalies, and which are artifacts of multiple testing. This limitation creates a bridge to later robust, Bayesian, and machine-learning approaches, all of which ask whether factor signals can survive estimation error, out-of-sample testing, and implementation costs.

4.3 | Bayesian Portfolio Theory and Black–Litterman: From Raw Expected Returns to Posterior Views

The Bayesian response to estimation risk occupies a distinct position between classical mean–variance theory and robust optimization. Mean–variance optimization is fragile because the expected-return vector is difficult to estimate. Small errors in μ can generate large and unstable changes in optimal weights. Bayesian portfolio theory addresses this problem by combining sample information with prior beliefs, equilibrium restrictions, or model uncertainty. Klein and Bawa (1976) show that estimation risk affects optimal portfolio choice, while Jorion (1986) uses Bayes–Stein shrinkage to stabilize expected-return estimates. Pástor (2000), Pástor and Stambaugh (2000), and Avramov and Zhou (2010) further develop Bayesian perspectives on portfolio choice, asset-pricing models, and model uncertainty.

Black–Litterman is the most important practitioner-facing version of this Bayesian logic. It begins with market-implied equilibrium returns and then incorporates investor views. The equilibrium-implied excess-return vector can be written as:

$$\pi = \delta \Sigma w_m,$$

where w_m is the market-capitalization portfolio and δ is a risk-aversion coefficient. Investor views are expressed as:

$$P\mu = Q + \varepsilon, \varepsilon \sim N(0, \Omega),$$

where P maps assets into views, Q is the vector of views, and Ω represents view uncertainty. The posterior expected-return vector is:

$$\mu_{BL} = \left[(\tau \Sigma)^{-1} + P' \Omega^{-1} P \right]^{-1} \left[(\tau \Sigma)^{-1} \pi + P' \Omega^{-1} Q \right].$$

Black and Litterman (1992) therefore change the role of expected returns. In classical Markowitz optimization, expected returns are direct inputs. In Black–Litterman, they are posterior outputs generated from equilibrium returns and investor views. Satchell and Scowcroft (2000) explain the Black–Litterman model as Bayesian portfolio construction and emphasize its relevance for financial management. He and Litterman (2002), Idzorek (2004), and Meucci (2010) further clarify implementation, intuition, confidence calibration, and extensions.

This literature is important for this survey because it directly addresses expected-return uncertainty, one of the central weaknesses of the Markowitz framework. Black–Litterman is not merely an implementation trick. It is a bridge between CAPM-style equilibrium, Bayesian updating, and institutional asset allocation. Its unresolved issues are prior sensitivity, view specification, confidence calibration, and governance. These issues also make Black–Litterman naturally connected to modern AI and LLM applications, where analyst views, textual signals, or macro narratives may be translated into structured portfolio views only if uncertainty and confidence are explicitly controlled.

4.4 | Downside Risk, VaR, and CVaR: From Variance to Tail Severity

The downside-risk literature begins from the observation that variance is not how many investors experience risk. Roy's safety-first approach prioritizes avoiding unacceptable outcomes, while Bawa (1975) and Fishburn (1977) formalize lower partial moments and below-target risk. The general lower partial moment is:

$$LPM_n(\tau) = E[\max(\tau - R_p, 0)^n]$$

When $n = 0$, it measures the probability of shortfall. When $n = 1$, it measures average shortfall magnitude. When $n = 2$, it becomes related to semivariance. Harlow and Rao (1989) generalize asset pricing in a mean-lower-partial-moment framework, while Harlow (1991) and Sortino and van der Meer (1991) translate downside risk into asset allocation and performance evaluation. The Sortino ratio can be written as:

$$\text{Sortino} = \frac{E(R_p - \tau)}{\sqrt{E[\min(R_p - \tau, 0)^2]}}$$

Downside risk matters because institutional objectives are often target-based. Pension funds care about liabilities. Endowments care about spending rules. Hedge funds care about drawdowns and redemption risk. Wealth managers care about capital preservation. This is why downside-risk measures are more intuitive for many real investors than total volatility. The limitation is that downside-risk models still require a target and may not fully measure the severity of extreme losses. This creates the transition to VaR, CVaR, and expected shortfall.

VaR became popular because it gives a simple answer to a regulatory and managerial question: how much can be lost over a horizon with a given confidence level? It is defined as:

$$\text{VaR}_\alpha(L) = \inf \{l : P(L \leq l) \geq \alpha\}.$$

However, VaR is not generally subadditive and does not quantify losses beyond the threshold. Artzner et al. (1999) responded by defining coherent risk measures through desirable axioms. Acerbi and Tasche (2002) further show why expected shortfall can be a coherent remedy to VaR's deficiencies.

CVaR enters portfolio theory because Rockafellar and Uryasev (2000, 2002) made tail risk optimizable. In sample form, CVaR

minimization can be written as:

$$\min_{w, \eta, u_i} \eta + \frac{1}{(1 - \alpha)N} \sum_{i=1}^N u_i$$

subject to:

$$u_i \geq L_i(w) - \eta, u_i \geq 0, 1'w = 1$$

This formulation allows CVaR to be embedded in linear programming and scenario optimization. Its contribution is not merely risk measurement; it changes the portfolio objective. A CVaR investor does not simply want low average variance. The investor wants low expected loss conditional on being in the worst tail of the distribution. This is especially relevant for derivatives, leverage, hedge funds, credit portfolios, and strategies with negative skewness.

The unresolved issue is scenario dependence. Historical samples may not contain enough crises; simulated scenarios may be misspecified; and tail correlations may change across regimes. Therefore, CVaR naturally connects with robust optimization, stress testing, volatility management, and regime-aware allocation.

4.5 | Robust Optimization and the 1/N Challenge: From Point Estimates to Input Ambiguity

The input-error problem is arguably the most important practical critique of classical optimization. Michaud (1989) argued that optimized portfolios can magnify estimation errors. Best and Grauer (1991) and Chopra and Ziemba (1993) showed that errors in expected returns, variances, and covariances can severely distort optimal portfolios. Jagannathan and Ma (2003) showed that long-only constraints can reduce risk even when they are theoretically “wrong,” because constraints act like implicit shrinkage. Ledoit and Wolf (2004) argued that the sample covariance matrix contains precisely the type of estimation error that perturbs a mean–variance optimizer. Kan and Zhou (2007) formalize optimal portfolio choice under parameter uncertainty.

DeMiguel et al. (2009) then disciplined the field by showing that optimized models usually fail to beat the naïve 1/N portfolio out of sample. Feng (2026) extends this debate to factor timing and volatility management, showing that even strategies designed to exploit time variation in factor returns or volatility often fail to outperform the 1/N benchmark. This result is important because 1/N contains no expected-return forecast, no covariance inversion, and no sophisticated optimization. It therefore becomes a benchmark for whether complexity adds real economic value or merely amplifies estimation noise. Kritzman et al. (2010) defend optimization against an overly broad interpretation of the 1/N result, but their response reinforces the same central point: optimization must be judged by out-of-sample value net of estimation error, turnover, and implementation costs.

Robust portfolio optimization formalizes the response to input ambiguity. Instead of trusting a single estimate of μ and Σ , the

investor optimizes against a plausible set of parameters:

$$\max_w \min_{\mu \in \mathcal{U}_\mu, \Sigma \in \mathcal{U}_\Sigma} \left(w' \mu - \frac{\gamma}{2} w' \Sigma w \right).$$

Here, $\mathcal{U}_\mu$ and $\mathcal{U}_\Sigma$ represent uncertainty sets for expected returns and covariances. Goldfarb and Iyengar (2003) develop robust portfolio-selection models with uncertainty structures, while Tütüncü and Koenig (2004) analyze robust allocation under worst-case behavior. Fabozzi et al. (2007) further help connect robust optimization with financial implementation.

This literature should be distinguished from Bayesian portfolio theory. Bayesian methods produce posterior expected returns by combining priors, equilibrium information, and investor views. Robust optimization instead asks what portfolio performs acceptably under adverse parameter realizations. Both respond to parameter uncertainty, but they do so differently: Black–Litterman modifies the expected-return input; robust optimization modifies the optimization problem itself.

4.6 | Risk Parity and Volatility-Managed Portfolios: From Return Optimization to Risk Control

Risk parity changes the object of diversification. In a traditional 60/40 portfolio, most risk may still come from equities. Risk parity instead asks each asset or asset class to contribute equally to total portfolio risk. If portfolio volatility is:

$$\sigma_p = \sqrt{w' \Sigma w},$$

then the marginal risk contribution of asset i is:

$$MR C_i = \frac{(\Sigma w)_i}{\sigma_p},$$

and total risk contribution is:

$$R C = w_i MRC_i.$$

Risk parity seeks:

$$R C_i = R C_j \text{ for all } i, j.$$

Qian (2005) presents risk parity as a framework for true diversification. Maillard et al. (2010) formalize equally weighted risk contribution portfolios, while Asness et al. (2012) link risk parity to leverage aversion in FAJ. Clarke et al. (2013) compare risk parity with maximum diversification and minimum variance, and López de Prado (2016) proposes hierarchical risk parity in JPM, using clustering and graph-based structure to address instability, concentration, and out-of-sample underperformance.

Risk parity is important for a broad survey because it connects Markowitz, robust estimation, and machine learning. It still depends on covariance, but it reduces reliance on expected-return estimates. It also anticipates ML-style clustering through hierarchical risk parity. Its weakness is that it may require

leverage, can be vulnerable to correlation breakdowns, and may fail if low-volatility assets become expensive or crowded.

Volatility-managed portfolios are the dynamic branch of this same risk-control logic. Volatility management begins from the empirical fact that volatility is more predictable than expected returns. Fleming et al. (2001) show that volatility timing has economic value. Barroso and Santa-Clara (2015) show that momentum risk is highly variable and predictable, and that risk-managed momentum can nearly eliminate crashes and almost double the Sharpe ratio. Daniel and Moskowitz (2016) show that momentum crashes occur in panic states after market declines, when volatility is high, and markets rebound. Moreira and Muir (2017) scale factor returns by inverse lagged realized variance and show large Sharpe-ratio improvements. A simple volatility-managed return is:

$$R_{t+1}^{VM} = \frac{c}{\hat{\sigma}_t^2} R_{t+1},$$

where $\hat{\sigma}_t^2$ is a forecast of conditional variance and c controls average exposure. The key contribution is that portfolio weights should not be static when risk is predictably time-varying. The limitation is that volatility scaling does not by itself identify regimes, expected-return changes, liquidity constraints, transaction costs, or crowded trades.

4.7 | Regime-Switching and State-Dependent Allocation: From Risk Scaling to State Probabilities

Regime-switching allocation extends dynamic portfolio theory beyond volatility scaling. Volatility management changes exposure as a function of conditional risk. Regime-switching allocation allows the entire investment opportunity set to vary across macro-financial states. Expected returns, volatilities, correlations, factor premia, and tail dependence may all differ across recession, expansion, crisis, recovery, inflationary, and low-volatility regimes.

Campbell and Viceira (1999) show that time-varying expected returns have important implications for long-horizon portfolio choice. Ang and Bekaert (2002) study international asset allocation under regime shifts. Guidolin and Timmermann (2007) identify crash, slow-growth, bull, and recovery states and show that optimal allocations vary considerably as investors revise state probabilities. In a state-dependent framework, portfolio weights can be written as:

$$w_t = \sum_{s=1}^S p_{t,s} w_s,$$

where $p_{t,s}$ is the probability of regime s at time t , and w_s is the allocation conditional on that regime.

This framework is important because diversification is itself state-dependent. Correlations and tail dependence often rise during crises, exactly when investors most need diversification. A static efficient frontier can therefore overstate diversification benefits in stressed markets. The unresolved issue is real-time identification. Regimes are often easier to identify ex post than ex ante. Regime-

switching allocation therefore connects naturally to machine learning, which can use macro variables, market states, volatility measures, text signals, and nonlinear features to improve state detection.

4.8 | Parametric Portfolio Policies and Machine-Learning Allocation: From Linear Factors to High-Dimensional Prediction

Machine learning enters portfolio theory through the information set. Classical models estimate μ and Σ from relatively low-dimensional historical data. Machine-learning models use high-dimensional firm characteristics, macro variables, factor signals, text, alternative data, nonlinear interactions, and time-varying states. Before the recent machine-learning literature, Brandt et al. (2009) offered an important data-driven bridge by proposing parametric portfolio policies that map firm characteristics and conditioning variables directly into portfolio weights. Rather than first estimating expected returns and then optimizing, their approach models the portfolio policy itself:

$$w_{i,t} = \bar{w}_{i,t} + \frac{1}{N_t} \theta' x_{i,t},$$

where $w_{i,t}$ and $\bar{w}_{i,t}$ are the final weight assigned to asset and the asset's weight in a predetermined benchmark portfolio separately. $x_{i,t}$ and θ are vector of standardized asset characteristics and vector of policy coefficients. The second term in the equation therefore represents a characteristic based overweight or underweight relative to benchmark. The characteristics are asset level signals, such as size, momentum, volatility, liquidity, profitability, or estimated factor exposures. They should not be interpreted as realized factor returns such as SMB, HML, MOM. Parametric portfolio policy is therefore different from factor based MVO, although these two terms are usually used in a mixed way: factor MVO optimizes across factor portfolios, whereas parametric portfolio policy uses characteristics observed for many individual assets to generate their weights directly.

In the original Brandt et al. (2009) formulation, the policy coefficients are estimated by maximizing the investor's sample average utility:

$$\hat{\theta} = \arg \max_{\theta} \frac{1}{T} \sum_{t=0}^{T-1} u(R_{p,t+1}(\theta)).$$

This utility maximization objective is specific to the original formulation rather than a universal requirement of all direct portfolio policy models. More generally, the same policy structure can be estimated under historical portfolio return, mean variance, Sharpe ratio, tail risk, or any other regularized objectives. The key information is that a large asset weight problem is reduced to the estimation of a relatively small vector θ , making parametric portfolio policies an important bridge to machine learning allocation.

The modern machine-learning literature expands this idea. Gu et al. (2020) compare machine-learning methods for measuring risk premia and show that trees and neural networks generate large economic gains by capturing nonlinear predictor interac-

tions. Freyberger et al. (2020) use nonparametric methods to identify which characteristics add incremental information and show the importance of nonlinearities. Feng et al. (2020) propose a factor-zoo testing method that evaluates new factors beyond a high-dimensional set of existing factors. Kelly et al. (2019) connect characteristics to covariances using IPCA, while Chen et al. (2024) estimate no-arbitrage asset-pricing models with deep neural networks. Bryzgalova et al. (2025) use decision trees to build interpretable cross-sections of stock returns and tradable test assets, showing that ML can support interpretability rather than simply produce black-box predictions.

The ML portfolio pipeline can be summarized as:

$$X_t \rightarrow f_{\theta}(X_t) \rightarrow \hat{\mu}_{t+1}, \hat{\Sigma}_{t+1}, \widehat{TailRisk}_{t+1} \rightarrow w_{t+1}$$

ML does not replace portfolio theory. It changes the estimation layer. The old questions remain: how should risk be measured, how should constraints be imposed, how should transaction costs be handled, how should out-of-sample performance be evaluated, and how should the resulting portfolio be governed? Without risk controls, explainability, and implementation discipline, machine learning can simply become a more flexible way to overfit.

4.9 | LLMs and Agent Investing: From Prediction to AI-Augmented Decision Systems

The newest literature moves from ML prediction to AI-augmented decision systems. Lopez-Lira and Tang (2026) show that ChatGPT scores from news headlines predict out-of-sample daily stock returns, especially for smaller stocks and negative news. FinGPT proposes an open-source financial LLM framework and emphasizes data curation, lightweight adaptation, robo-advising, algorithmic trading, and low-code development. FinMem introduces an LLM trading agent with layered memory and character design, explicitly moving from text interpretation to agentic financial decision-making. Kong et al. (2024) in JPM discuss LLM applications and benchmarks for financial and investment management, making JPM one of the first practical outlets to integrate LLMs into the portfolio-management discussion.

The LLM/agent frontier can be written as:

$$\begin{aligned} &Data \rightarrow Retrieval \rightarrow Reasoning \rightarrow Risk Filter \\ &\rightarrow Trade or Allocation Action. \end{aligned}$$

This is not yet a mature portfolio theory. It is an emerging layer on top of the earlier field. The key unresolved issues are hallucination, prompt sensitivity, market impact, backtest leakage, concentration, overfitting, weak auditability, and governance. Therefore, the correct research agenda is not "LLMs replace portfolio managers." It is "LLMs and agents must be embedded inside robust, tail-aware, constraint-aware, explainable portfolio systems."

Figure 9 illustrates how machine learning and AI enter the portfolio-management pipeline. The figure separates the data and signal layer, intelligence and prediction layer, portfolio construc-

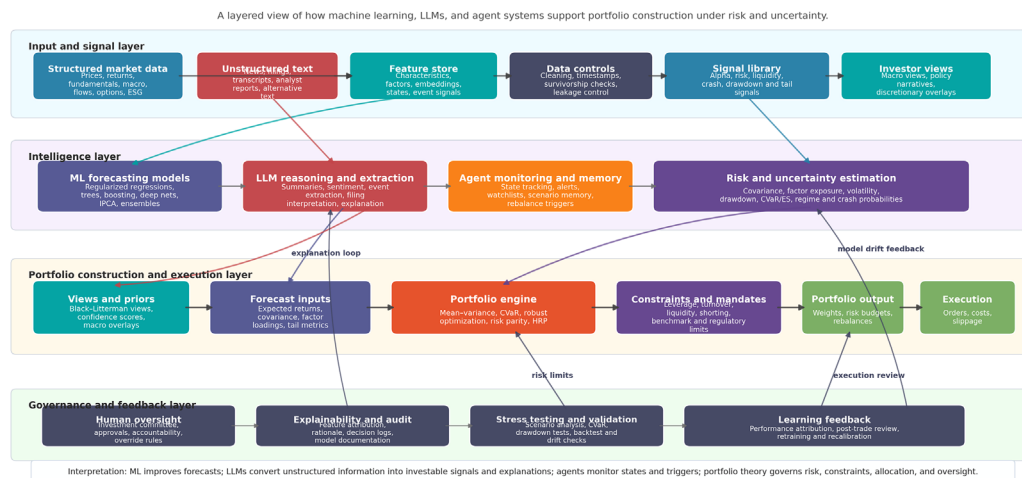


FIGURE 9 | ML/AI-augmented portfolio management architecture. The figure illustrates how machine learning, large language models, and agent-based systems can support portfolio management across four layers: data and signals, intelligence and prediction, portfolio construction and execution, and governance and feedback. It emphasizes that AI does not replace portfolio theory; rather, it assists forecasting, information extraction, monitoring, and explanation, while portfolio theory continues to govern optimization, risk constraints, execution discipline, and human oversight. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

tion and execution layer, and governance and feedback layer. This distinction is important because AI does not replace portfolio theory at a single point. Instead, it can assist multiple stages of the investment process, from extracting signals to generating views, monitoring risks, explaining decisions, and enforcing constraints. Figure 9 also clarifies the boundary between prediction and portfolio management. Machine learning can improve forecasts of expected returns, volatility, factor exposures, regimes, and tail risk, while LLMs can process news, filings, analyst reports, and macro narratives. However, the final portfolio still requires a disciplined allocation engine, transaction-cost control, CVAR or drawdown constraints, robust uncertainty sets, and human governance. This is why AI-augmented portfolio management should be interpreted as an extension of portfolio theory rather than a replacement for it.

The AI and LLM frontier therefore, should not be interpreted as a break from portfolio theory. It is better understood as an additional information-processing and decision-support layer on top of the earlier theoretical architecture. Machine learning can improve forecasts of returns, volatility, factor exposures, regimes, and tail risk, while LLMs can process news, filings, analyst reports, and macro narratives. However, portfolio construction still requires the core elements developed by previous generations: diversification, factor exposure control, Bayesian updating, Black–Litterman views, downside-risk preferences, CVAR constraints, robust parameter treatment, risk-budgeting discipline, transaction-cost control, and governance. AI-augmented portfolio management is therefore most promising when it is embedded within, rather than detached from, the cumulative structure of portfolio theory.

Table 5 summarizes the main papers discussed in the thematic review. The purpose of the table is not to provide a complete bibliography, but to show how representative studies contribute to different generations of portfolio theory. Table 5 shows that the field develops through linked conceptual steps. Roy and Markowitz define the initial risk-choice problem; Sharpe and

Fama–French connect portfolio choice to systematic risk; Jorion and Black–Litterman address expected-return uncertainty; Bawa, Fishburn, Harlow, and Sortino shift attention to downside risk; Artzner and Rockafellar–Uryasev formalize coherent and tail-risk optimization; Michaud, Ledoit–Wolf, and DeMiguel et al. expose the input-error problem; Maillard et al. and López de Prado shift attention to risk budgeting; Moreira–Muir and Guidolin–Timmermann introduce conditional allocation; Brandt et al. connect characteristics directly to portfolio weights; and Gu–Kelly–Xiu, Chen–Pelger–Zhu, and recent LLM studies move the field toward AI-augmented investment systems.

The thematic review points to several future research opportunities, summarized in Table 6. These opportunities are organized around the unresolved tensions that recur throughout the survey: tail risk, expected-return uncertainty, parameter ambiguity, regime dependence, model explainability, institutional constraints, and human–AI governance. Table 6 suggests that the next generation of portfolio theory should be integrative. Future work should not treat Bayesian views, ML forecasts, CVAR constraints, robust optimization, risk parity, and LLM agents as separate literatures. The more promising direction is to combine them into portfolio systems that are predictive, tail-aware, robust to estimation error, regime-sensitive, explainable, and implementable under real investment constraints.

5 | Discussion

The central message of this survey is that portfolio theory has evolved through successive expansions of the meaning of uncertainty. Markowitz (1952) defined the modern portfolio problem by showing that risk is not the variance of individual assets but the covariance structure of the portfolio. This insight remains the foundation of the field because nearly all later models, including factor allocation, robust optimization, risk parity, volatility-managed portfolios, and machine-learning allocation, still depend on how assets co-move. However, the history reviewed in

TABLE 5 | Selected paper summary and context.

Study	Outlet	Generation	Main contribution
Roy (1952)	Econometrica	Safety-first	Defines risk as probability of disaster
Markowitz (1952)	JF	Mean–variance	Formalizes efficient diversification
Sharpe (1964)	JF	CAPM	Prices systematic market risk
Fama and French (1993)	JFE	Factors	Establishes common stock and bond factors
Bawa (1975)	JFE	Downside risk	Develops partial-moment ordering
Fishburn (1977)	AER	Downside risk	Formalizes below-target risk
Jorion (1986)	JFQA	Bayesian/shrinkage	Stabilizes expected-return estimates through Bayes–Stein shrinkage
Harlow and Rao (1989)	JFQA	Downside asset pricing	Generalizes mean-LPM asset pricing
Michaud (1989)	FAJ	Estimation risk	Critiques optimization as error amplification
Black and Litterman (1992)	FAJ	Bayesian allocation	Combines equilibrium-implied returns and investor views
Artzner et al. (1999)	Mathematical Finance	Coherent risk	Defines coherent risk measures
Rockafellar and Uryasev (2000)	Journal of Risk	CVaR	Makes CVaR optimization tractable
Satchell and Scowcroft (2000)	JAM	Black–Litterman	Explains Bayesian allocation in practice
Ledoit and Wolf (2004)	JPM	Shrinkage	Stabilizes covariance estimation
Idzorek (2004)	Ibbotson Associates	Black–Litterman implementation	Incorporates user-specified confidence levels
DeMiguel et al. (2009)	RFS	Benchmarking	Shows difficulty of beating 1/N
Feng (2026)	FMPM	Volatility management/factor MVO/1/N benchmark	Shows difficulty of beating 1/N
Brandt et al. (2009)	RFS	Parametric portfolio policies	Maps characteristics directly into portfolio weights
Maillard et al. (2010)	JPM	Risk parity	Formalizes equal risk contribution
Moreira and Muir (2017)	JF	Volatility management	Scales factor exposure by volatility
Koumou (2020)	FMPM	Survey/risk parity	Reviews diversification principles
Gu et al. (2020)	RFS	ML	Shows nonlinear ML gains in asset pricing
Chen et al. (2024)	Management Science	Deep learning	Builds no-arbitrage deep-learning model
Lopez-Lira and Tang (2026)	SSRN	LLM investing	Uses ChatGPT news scores for return prediction
Kong et al. (2024)	JPM	LLM investment management	Reviews LLM applications and benchmarks

this article also shows that mean–variance theory is incomplete when risk is asymmetric, parameters are uncertain, markets shift across regimes, and investors operate under constraints.

A particularly important implication is that estimation risk should not be treated only as a prelude to robust optimization. Bayesian portfolio theory and the Black–Litterman model represent a distinct response to parameter uncertainty because they change how expected returns enter the allocation prob-

lem. Instead of relying on noisy sample means, they combine equilibrium restrictions, priors, and investor views. This makes Black–Litterman a key bridge between Markowitz optimization, CAPM-style equilibrium, and institutional portfolio management.

This survey contributes to the literature by reframing portfolio theory as a sequence of expanding uncertainty concepts rather than as a collection of separate optimization techniques. This

TABLE 6 | Research agenda from the thematic synthesis.

Agenda	Main question	Suggested approach
Tail-aware ML portfolios	Can ML forecasts be optimized under CVaR constraints?	ML prediction + CVaR optimization
Bayesian and LLM-assisted views	Can analyst, macro, or LLM-generated views be incorporated safely?	Black–Litterman + confidence calibration + human review
Distributionally robust allocation	How can portfolios remain stable under unknown tail distributions?	Robust uncertainty sets + expected shortfall
Regime-conditioned risk parity	Can risk contribution be equalized across regimes, not only unconditionally?	Regime switching + risk budgeting
Explainable AI allocation	Which features drive AI-generated portfolio weights?	SHAP, decision trees, attribution constraints
Agentic portfolio governance	How should autonomous agents be constrained?	Risk filters, audit trails, human-in-the-loop rules
Public-data replicability	Can advanced allocation be tested with public data?	Fama–French, FRED, ETF, WRDS/CRSP benchmarks

framing helps reconcile literatures that are often reviewed separately: mean–variance theory, downside-risk models, VaR/CVaR optimization, robust optimization, risk parity, volatility-managed portfolios, regime-switching allocation, machine-learning asset pricing, and LLM-based investing. The resulting perspective is especially useful for economic surveys because it explains not only what methods exist, but why each generation emerged in response to the limitations of earlier models.

The first major extension is from total variance to systematic risk. Sharpe (1964), Lintner (1965), Mossin (1966), and Ross (1976) moved the field from individual portfolio choice to equilibrium asset pricing. In this generation, portfolio theory becomes a theory of priced risk: investors should be compensated for bearing non-diversifiable systematic exposure rather than idiosyncratic volatility. Fama and French (1992, 1993), Carhart (1997), Jegadeesh and Titman (1993), and Asness et al. (2013) then transformed the theory into factor allocation. This development remains central to asset management because institutional portfolios are often evaluated through exposures to market, size, value, momentum, profitability, investment, low-risk, and other factors.

The second extension is from symmetric variance to downside risk. Roy (1952), Bawa (1975), Fishburn (1977), Harlow and Rao (1989), Harlow (1991), and Sortino and van der Meer (1991) show that investors do not usually dislike upside volatility. They dislike below-target returns, funding shortfalls, drawdowns, and ruin. This matters especially for pensions, insurers, endowments, sovereign funds, hedge funds, and private clients. A variance-minimizing portfolio can still be unattractive if it exposes investors to unacceptable left-tail outcomes. Downside-risk theory therefore gives portfolio theory a more realistic investor-objective foundation.

The third extension is from downside risk to tail-risk optimization. VaR made tail exposure communicable, but Artzner et al. (1999) showed why risk measures should satisfy coherence properties such as subadditivity. Rockafellar and Uryasev (2000, 2002) made CVaR central because expected shortfall can be optimized

in a tractable way. This changed portfolio theory from controlling average dispersion to controlling expected losses in adverse states. In practice, CVaR is especially important for leveraged portfolios, derivative strategies, credit portfolios, momentum strategies, hedge funds, and institutions with drawdown-sensitive stakeholders.

The fourth extension is from return uncertainty to parameter uncertainty. Michaud (1989), Chopra and Ziemba (1993), Black and Litterman (1992), Jagannathan and Ma (2003), Ledoit and Wolf (2004), Kan and Zhou (2007), DeMiguel et al. (2009), and Tu and Zhou (2011) show that optimized portfolios can fail because inputs are estimated with error. This literature is crucial because it explains why mathematically sophisticated portfolios may perform poorly out of sample. Robust optimization, shrinkage, Bayesian views, constraints, and 1/N benchmarks are not minor technical details; they are responses to the central implementation problem of portfolio theory.

The fifth extension is from capital allocation to risk allocation. Risk parity and risk budgeting, developed in both academic and practitioner-oriented outlets, shift attention from asset weights to risk contributions. Maillard et al. (2010), Asness et al. (2012), and López de Prado (2016) show that diversification should be assessed through marginal and total risk contribution rather than nominal capital weights. This is why risk parity belongs in a broad survey of portfolio theory: it offers a practical response to the instability of expected-return estimates and the dominance of equity risk in traditional balanced portfolios.

The sixth extension is from static allocation to conditional allocation. Fleming et al. (2001), Barroso and Santa-Clara (2015), Daniel and Moskowitz (2016), and Moreira and Muir (2017) show that volatility timing and volatility-managed portfolios can improve risk-adjusted performance. Regime-switching allocation, including Ang and Bekaert (2002) and Guidolin and Timmermann (2007), extends this insight by allowing expected returns, volatilities, correlations, and optimal weights to differ across market states. This is important because diversification

benefits often decline during crises, precisely when investors need them most.

The final extension is from low-dimensional models to high-dimensional AI-assisted decision systems. Gu et al. (2020), Kelly et al. (2019), Freyberger et al. (2020), Feng et al. (2020), Chen et al. (2024), and Bryzgalova et al. (2025) show that machine learning can improve asset-pricing and portfolio inputs by capturing nonlinearities, interactions, and latent structures. More recent LLM and agent-investing studies, such as Lopez-Lira and Tang (2026), FinGPT, FinMem, and Kong et al. (2024), suggest that AI can assist not only prediction but also text processing, scenario reasoning, explanation, monitoring, and investment workflow automation.

However, AI does not remove the classical problems of portfolio theory. It may amplify them. Machine-learning models can overfit, learn unstable relations, generate opaque signals, ignore transaction costs, and produce fragile allocations. LLMs and agents introduce additional risks: hallucination, prompt sensitivity, data leakage, weak auditability, and unclear accountability. Therefore, the future of portfolio theory should not be framed as “AI replaces Markowitz.” A stronger agenda is “AI is embedded within a portfolio-theory architecture.” That architecture should combine diversification, factor structure, downside-risk preferences, CVaR tail control, robust uncertainty sets, risk budgeting, volatility scaling, regime conditioning, explainability, and human governance.

The continuing relevance of the $1/N$ benchmark is especially important for this AI frontier. If traditional factor-timing and volatility-managed strategies do not consistently dominate simple diversified rules, then AI-based allocation faces an even higher burden of out-of-sample, turnover-adjusted, and benchmark-relative validation (DeMiguel et al. 2009; Feng 2026). This benchmark discipline is not a rejection of sophisticated allocation. Rather, it is a necessary condition for evaluating whether additional modeling complexity creates genuine economic value after estimation error and implementation frictions are taken into account.

This also explains why practical journals such as FAJ, JPM, FMPM, and JAM are essential to the survey. Top journals often develop the theory, but practitioner outlets test whether those ideas survive implementation. The next generation of research should therefore bridge academic asset pricing, operations research, machine learning, LLM/AI, and institutional portfolio management.

6 | Conclusion

This article reviews the development of portfolio theory from mean–variance optimization to AI-augmented portfolio management. The main conclusion is that the field has evolved by repeatedly redefining uncertainty. Markowitz (1952) formalized risk as variance and covariance. CAPM and factor models shifted attention to systematic exposure. Bayesian portfolio theory and Black–Litterman addressed expected-return uncertainty by combining priors, equilibrium-implied returns, and investor views. Downside-risk models reframed risk as below-target shortfall.

VaR and CVaR made tail loss central. Robust optimization addressed ambiguity and worst-case parameter uncertainty. Risk parity changed diversification from capital weights to risk contributions. Volatility-managed and regime-switching portfolios introduced time-varying states. Machine learning and AI now expand the information set and allow nonlinear prediction, text processing, and decision support.

The review also shows that portfolio theory develops through interaction between academic and practitioner communities. Top journals such as JF, JFE, RFS, JFQA, RF, and Management Science provide theoretical and empirical breakthroughs. Practitioner-oriented journals such as FAJ, JPM, FMPM, and JAM translate these ideas into implementable allocation frameworks, risk controls, and institutional investment processes.

The future frontier is integrative. AI should not be treated as a replacement for portfolio theory, but as an additional layer within a disciplined investment architecture. Promising research should combine machine-learning forecasts, LLM-based information processing, CVaR constraints, robust optimization, regime awareness, transaction-cost control, explainability, and human oversight. In this sense, the next stage of portfolio theory is likely to be AI-augmented, but still rooted in the same core problem first formalized by Markowitz: how to allocate capital under risk and uncertainty.

Author Contribution:

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